

Harvineet Singh

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Summary

Data scientist specializing in healthcare-focused machine learning with expertise in causal inference, robust learning, and algorithmic fairness. Proven ability to analyze complex clinical datasets, rigorously evaluate predictive models, and diagnose performance drifts. Experience in collaborating with cross-functional teams to develop scalable ML systems for digital analytics applications. Passionate about leveraging data science and artificial intelligence to improve health outcomes and advance responsible AI in health technology.

Education

- 2018–2023 **New York University, Center for Data Science, USA,**
Ph.D. in DATA SCIENCE GPA: 4.0/4.0.
Thesis: [Fair, Robust, and Data-Efficient Machine Learning for Health](#)
Thesis committee: [Rumi Chunara](#), [Rajesh Ranganath](#), [Jennifer Hill](#), [Fei Wang](#), [Daniel B. Neill](#)
My dissertation focuses on understanding and designing principled **safeguards to deploy ML** in the real world. Recognizing challenges along the entire ML pipeline, my work integrates contextual knowledge with causal and statistical thinking to [collect better data](#), to develop [fair algorithms](#), and to [evaluate robustness](#) to data shifts.
- 2010–2015 **Indian Institute of Technology (IIT) Delhi, Department of Mathematics, India,**
Integrated Master of Technology in MATHEMATICS AND COMPUTING GPA: 8.7/10.0.
My dissertation applied **natural language processing** (NLP) methods in conjunction with graph theory and machine learning to online social network data for predicting [content virality](#) and [topic adoption](#).

Professional Experience

- 2025–Present **Qualified Health, USA.**
MEMBER OF TECHNICAL TEAM
 - Assessed the safety and accuracy of agentic workflows for data abstraction and classification of clinical notes.
 - Contributed to the development of a Python package to conduct multi-faceted evaluation of LLM workflows.
 - Wrote system cards for generative AI-based applications and reported validation results on clinical data.
- 2023–2025 **University of California, San Francisco, USA.**
POSTDOCTORAL SCHOLAR, Mentor: [Jean Feng](#)
 - Developed statistical tools to diagnose **performance drifts** in ML systems, ensuring robustness.
 - Evaluated ML models across multi-institutional datasets to ensure **safe and reliable** deployment.
 - Designed **data collection** strategies to measure bias in ML pipelines, aligning with responsible ML practices.
- 2015–2018 **Adobe Research, India.**
RESEARCH ENGINEER
 - Built **scalable** ML models for customer behavior analytics and **recommender systems**.
 - **Transferred multiple technologies** to Adobe Analytics collaborating with cross-functional teams.
 - Filed 6 **patents**, published 3 conference papers, showcased **product demos** at external conferences.

Research Experience

- Jun–Oct 2022 **Amazon AWS, Germany.**
APPLIED SCIENTIST INTERN, Mentors: [Chris Russell](#) and [Matthäus Kleindessner](#).
 - Conducted a large-scale **benchmarking study** of fairness methods across tabular datasets and algorithms.
 - Explored theoretical properties of tree-based models, focusing on achieving **minmax-fairness** in practice.
- Jun–Aug 2021 **Microsoft Research, USA.**
RESEARCH INTERN, Mentors: [Emre Kiciman](#) and [Adith Swaminathan](#).
 - Developed a method to optimize **contextual bandit** models as part of **ad ranking** pipelines.
 - Proposed a **causal feature importance** measure to enhance robustness of decision-making systems.

Jun–Aug **Harvard University**, *Center for Research on Computation and Society*, USA.

2020 SUMMER RESEARCH FELLOW, Mentor: [Hima Lakkaraju](#) and [Finale Doshi-Velez](#).

- Developed robust evaluation methods for **reinforcement learning** policies from offline data.
- Integrated **adversarial learning** with causal inference to evaluate policies in non-stationary environments.

May–Jul 2014 **Adobe Research**, India.

RESEARCH INTERN, Mentor: [Moumita Sinha](#).

- Proposed a machine learning model to predict **customer churn** from large web **clickstream data**.
- Work **productized** in Adobe Experience Cloud and validated on online **marketing data** (US 11,170,407).

Publications

- [Optimal Vasopressin Initiation in Septic Shock: The OVISS Reinforcement Learning Study](#).
Alexandre Kalimoultou, Jason N. Kennedy, Jean Feng, [Harvineet Singh](#), Suchi Saria, Derek C. Angus, Christopher W. Seymour, Romain Pirracchio.
Journal of the American Medical Association (**JAMA**) 2025.
- [A hierarchical decomposition for explaining ML performance discrepancies](#).
[Harvineet Singh](#), Fan Xia, Adarsh Subbaswamy, Alexej Gossmann, Jean Feng.
Conference on Neural Information Processing Systems (**NeurIPS**) 2024.
- [Monitoring the performance of machine learning algorithms that induce feedback loops: what is the causal estimand?](#)
Jean Feng, Adarsh Subbaswamy, Alexej Gossmann, [Harvineet Singh](#), Berkman Sahiner, Mi-Ok Kim, Gene Pennello, Nicholas Petrick, Romain Pirracchio, Fan Xia.
Causal Learning and Reasoning (**CLearR**) 2024.
- [When do Minimax-fair Learning and Empirical Risk Minimization Coincide?](#).
[Harvineet Singh](#), Matthäus Kleindessner, Volkan Cevher, Rumi Chunara, and Chris Russell.
International Conference on Machine Learning (**ICML**) 2023.
- [“Why did the Model Fail?”: Attributing Model Performance Changes to Distribution Shifts](#).
Haoran Zhang*, [Harvineet Singh](#)*, Marzyeh Ghassemi, and Shalmali Joshi.
International Conference on Machine Learning (**ICML**) 2023.
- [Harvineet Singh](#) and Rumi Chunara.
[Measures of Disparity and their Efficient Estimation](#).
Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society (**AIES**) 2023.
- [Data Poisoning Attacks on Off-Policy Policy Evaluation Methods](#).
Elita Lobo, [Harvineet Singh](#), Marek Petrik, Cynthia Rudin, and Himabindu Lakkaraju.
Conference on Uncertainty in Artificial Intelligence (**UAI**) 2022. **Oral Presentation**.
- [Towards Robust Off-Policy Evaluation via Human Inputs](#).
[Harvineet Singh](#), Shalmali Joshi, Finale Doshi-Velez, and Himabindu Lakkaraju.
AAAI/ACM Conference on AI, Ethics, and Society (**AIES**) 2022.
- [Generalizability challenges of mortality risk prediction models: A retrospective analysis on a multi-center database](#).
[Harvineet Singh](#), Vishwali Mhasawade, and Rumi Chunara.
PLOS Digital Health 2022.
- [Fairness Violations and Mitigation under Covariate Shift](#).
[Harvineet Singh](#), Rina Singh, Vishwali Mhasawade, and Rumi Chunara.
ACM Conference on Fairness, Accountability, and Transparency (**FAccT**) 2021.
- [Cascading Linear Submodular Bandits: Accounting for Position Bias and Diversity in Online Learning to Rank](#).
Gaurush Hiranandani*, [Harvineet Singh](#)*, Prakhar Gupta*, Iftikhar A. Burhanuddin, Zheng Wen, and Branislav Kveton.
Conference on Uncertainty in Artificial Intelligence, (**UAI**) 2019. **Oral Presentation**.
- [Stuck? No Worries!: Task-aware Command Recommendation and Proactive Help for Analysts](#).
Aadhavan M. Nambhi*, Bhanu P. R. Guda*, Aarsh P. Agarwal*, Gaurav Verma, [Harvineet Singh](#) and Iftikhar A. Burhanuddin.
ACM International Conference on User Modeling, Adaptation, and Personalization, (**UMAP**) 2019.

- [Modeling Time to Open of Emails with a Latent State for User Engagement Level](#).
Moumita Sinha, Vishwa Vinay, and Harvineet Singh.
ACM International Conference on Web Search and Data Mining, (**WSDM**) 2018.
- [Modeling Hint-Taking Behavior and Knowledge State of Students with Multi-Task Learning](#).
Ritwick Chaudhry*, [Harvineet Singh*](#), Pradeep Dogga, and Shiv Saini.
International Conference on Educational Data Mining, (**EDM**) 2018.
- [Show and Recall: Learning What Makes Videos Memorable](#).
Sumit Shekhar, Dhruv Singal, [Harvineet Singh](#), Manav Kedia, and Akhil Shetty.
Workshop on MBCC at IEEE International Conference on Computer Vision (**ICCV**) 2017.
- [On the role of conductance, geography and topology in predicting hashtag virality](#).
Siddharth Bora, [Harvineet Singh](#), Anirban Sen, Amitabha Bagchi, and Parag Singla.
Springer Journal on Social Network Analysis and Mining (**SNAM**) 2015.

Patents

- Natural-language based Intelligent Analytics Interface. (US 11,321,373)
- Summarizing Video Content based on Memorability of the Video Content. (US 10,311,913)
- Personalized E-Learning Using A Deep-Learning-Based Knowledge Tracing And Hint-Taking Propensity Model. (US 10,943,497)
- Predicting Unsubscription of Subscribing Users. (US 11,170,407)
- End of period metric projection with intra-period alerts. (US 11,205,111)
- Intelligent customer journey mining and mapping. (US 11,756,058)

Key Skills

Languages	Python, R, SQL (proficient); Java, JavaScript (familiar).
Methods	Model development and monitoring, robust optimization, deep learning, LLM evaluation
Tools	PyTorch, JAX, TensorFlow, Apache Spark, Scikit-Learn, Git, AWS Cloud.

Awards

2022	Selected for Future Leaders Summit on Responsible Data Science at University of Michigan.
2019	Awarded Travel Grants to attend UAI 2019, and FairML4H@NeurIPS 2019.
2018-2023	Awarded Graduate Research Fellowship by NYU Center for Data Science.
2014	Awarded Scholarship by Ministry of Human Resource Development for academic excellence.
2010-2012	IITD Semester Merit Award for meritorious academic performance (top 7% of batch).

Professional Service

2023	General Chair for the Machine Learning for Health Symposium 2023.
2022	Finance Co-chair for the Machine Learning for Health Symposium 2022.
2022	Tutorial Co-chair for the Conference on Health, Inference, and Learning 2022.
2022	Student Co-organizer Trustworthy AI Day at KDD 2022.
2022	Co-organizer NYU AI School , a week-long school for undergraduates interested in AI.
2021	Student Organizer for Deep Learning Day at KDD 2021.
2020–2021	Co-lead Causal inference and fairness group in Mechanism Design for Social Good (MD4SG).
2020	Research Mentor Miao Zhang (Ph.D. at NYU), Lazarus Chok (M.S. at NYU)
	Paper Reviewer for Statistics and CS venues: Statistics in Medicine, JRSS Series C, JNCI Cancer Spectrum, NeurIPS, ICLR (Highlighted Reviewer), ICML, AISTATS, FAccT, AIES, MLHC, CHIL, ML4H, EAAMO.

Press Coverage of Work

May 17, 2022 [Finding the Fairness in AI](#). **Communications of ACM**.

- May 11, 2022 [Responsible Data Science and AI: Accounting for Dataset Shifts and Designing Fair Machine Learning Systems.](#) **NYU Center for Data Science Blog.**
- Feb 25, 2020 [Researchers Ask: Will You Remember that Video Tomorrow?](#) **Adobe Research Blog.**
- Aug 29, 2017 [Adobe wants to bring its AI smarts to email marketing campaigns.](#) **TechCrunch.**
- Mar 25, 2016 [Adobe Previews In-Development Features at Summit 2016.](#) **Techvibes.**

Recent Talks

- Nov, 2024 'Explaining ML performance changes across hospital sites' at AMIA Annual Symposium.
- Aug, 2024 'A framework to explain performance changes of ML models' at Joint Statistical Meetings.
- Jul, 2024 'Explaining performance changes of ML models across populations' at UCSF BCHSI meeting.
- May, 2024 'Machine learning for causal inference' guest lecture in Advanced ML course at UCSF.
- Mar, 2024 'Fair and robust learning' in Fair and Explainable Decision Making lab at UMass Amherst.
- Dec, 2023 'Sample size calculations for fairness audits' at Regulatable ML workshop at NeurIPS.
- Nov, 2023 'Fair machine learning in health' at Future of AI in Medicine seminar at UCSF.
- Oct, 2023 'Responsible machine learning for health' guest lecture in a course at Columbia University.
- July, 2023 'Fair and reliable machine learning for health' at McKinsey AI seminar.